

Distributed System

BCA — Semester VI — 2025 — Answer Sheet
Subject Code: CACS352

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. "Distributed system acts as a single coherent system to its end user." Justify the statement with its features and challenges.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate between client server and peer to peer architecture in distributed system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Describe transient synchronous communication with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain wrapper and interceptor in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Describe the working mechanism of distributed commit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Discuss the different attacks that can target distributed systems and explain how they can be mitigated.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short note (Any two) a) DNS Query Types b) Checkpointing c) Digital Signature

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions [2x10=20]

9. What do you mean by logical clock? Explain Lamport's Logical clock.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How can you claim that replication is one of the scaling techniques in distributed system? Explain consistency protocols with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is virtualization? Explain different types of virtualizations with example. Explain the function of middleware in distributed system with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.